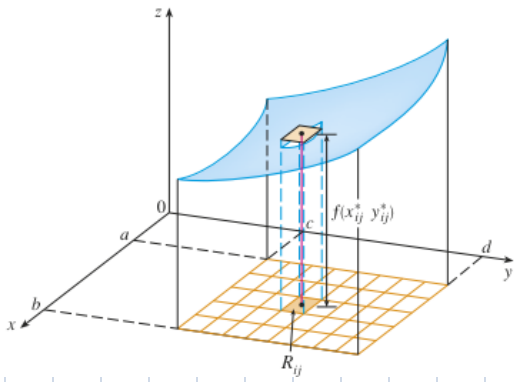




Recap: $f: \mathbb{R}^n \rightarrow \mathbb{R}$

$$f(x_1, x_2, \dots, x_n)$$

$$f(x_1 + \Delta x_1, x_2 + \Delta x_2, \dots, x_n + \Delta x_n) \approx f(x_1, x_2, \dots, x_n) + \Delta x_1 \frac{\partial f}{\partial x_1} + \Delta x_2 \frac{\partial f}{\partial x_2} + \dots$$

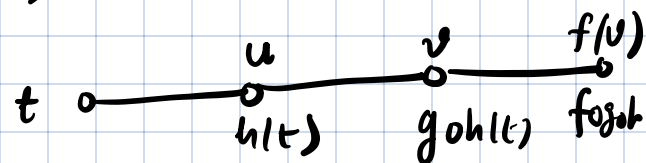
Chain rule: $f(g(t))$

$$(f \circ g)'(t) = f'(g(t)) g'(t)$$

$$(f \circ g \circ h)'(t) = f'(g(h(t))) g'(h(t)) h'(t)$$

$$u = h(t)$$

$$v = g(h(t))$$



Consider $\underbrace{f(x, y)}$ where $x = g_1(t)$ & $y = g_2(t)$ are functions of t .

$$f: \mathbb{R}^2 \rightarrow \mathbb{R}$$

$\underbrace{f(g_1(t), g_2(t))}$ is a one variable fn $\mathbb{R} \rightarrow \mathbb{R}$

$$\frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}$$

Can be seen from linear approx.

$$f(x+\Delta x, y+\Delta y) \approx f(x, y) + \frac{\partial f}{\partial x} \Delta x + \frac{\partial f}{\partial y} \Delta y$$

Using $\Delta x \approx \frac{dx}{dt} \Delta t$, $\Delta y = \frac{dy}{dt} \Delta t$

we get

$$f(x+\Delta x, y+\Delta y) \approx f(x, y) + \Delta t \left(\frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} \right)$$

In general $f(x_1, x_2, \dots, x_n)$ where $x_i = g_i(t)$

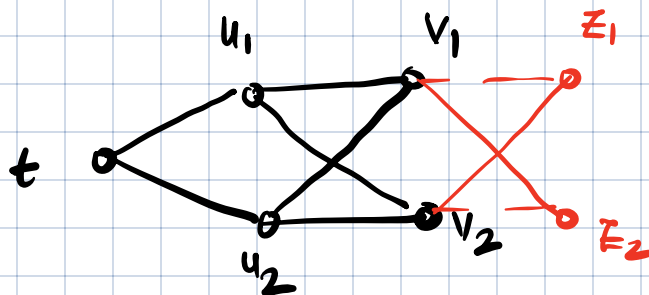
then the derivative is $\sum \frac{\partial f}{\partial x_i} \frac{dx_i}{dt}$

Example:

$$x^2 y + 3xy^4$$

$$x = \sin t, \quad \cos t$$

$$\underbrace{(2xy + 3y^4)}_{\frac{\partial f}{\partial x}} \underbrace{\cos t}_{\frac{dx}{dt}} + \underbrace{(x^2 + 12xy^3)}_{\frac{\partial f}{\partial y}} \underbrace{(-\sin t)}_{\frac{dy}{dt}}$$



$$u_1 = g_1(t)$$

$$u_2 = g_2(t)$$

$$v_1 = f_1(u_1, u_2)$$

$$v_2 = f_2(u_1, u_2)$$

$$\frac{dv_1}{dt} = \frac{\partial v_1}{\partial u_1} \frac{du_1}{dt} + \frac{\partial v_1}{\partial u_2} \frac{du_2}{dt}$$

$$\frac{dv_2}{dt} = \frac{\partial v_2}{\partial u_1} \frac{du_1}{dt} + \frac{\partial v_2}{\partial u_2} \frac{du_2}{dt}$$

$$\begin{bmatrix} \frac{dv_1}{dt} \\ \frac{dv_2}{dt} \end{bmatrix} = \begin{bmatrix} \frac{\partial v_1}{\partial u_1} & \frac{\partial v_1}{\partial u_2} \\ \frac{\partial v_2}{\partial u_1} & \frac{\partial v_2}{\partial u_2} \end{bmatrix} \begin{bmatrix} \frac{du_1}{dt} \\ \frac{du_2}{dt} \end{bmatrix}$$

(Back-propagation)

$$\frac{\partial z_1}{\partial t} = \underbrace{\frac{\partial z_1}{\partial v_1}}_{\text{computed in the recursive step}} \underbrace{\frac{dv_1}{dt}}_{\text{computed in the recursive step}} + \underbrace{\frac{\partial z_1}{\partial v_2}}_{\text{computed in the recursive step}} \underbrace{\frac{dv_2}{dt}}_{\text{computed in the recursive step}} \} \text{ implemented by a recursive function call}$$

Vector valued functions of multiple variables

$$f: \mathbb{R}^n \rightarrow \mathbb{R}^m$$

$$f: \mathbb{R}^n \rightarrow \mathbb{R}$$

$$\nabla f$$

$$\begin{bmatrix} f_1(x_1, x_2, \dots, x_n) \\ f_2(x_1, x_2, \dots, x_n) \\ f_3(x_1, x_2, \dots, x_n) \\ \vdots \\ f_m(x_1, x_2, \dots, x_n) \end{bmatrix}$$

A change of $\underline{\Delta x}$ in $(x_1, x_2, \dots, x_n)$ results in a change of $\nabla f_i^T \underline{\Delta x}$ in f_i for $i=1, 2, \dots, m$.

Change in f_i

$$\begin{bmatrix} \Delta f_1 \\ \Delta f_2 \\ \vdots \\ \Delta f_m \end{bmatrix} = \begin{bmatrix} \nabla f_1^T \\ \nabla f_2^T \\ \vdots \\ \nabla f_m^T \end{bmatrix} \begin{bmatrix} \Delta x_1 \\ \Delta x_2 \\ \vdots \\ \Delta x_n \end{bmatrix}$$

Change in f_m

$$= \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \dots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \dots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \frac{\partial f_m}{\partial x_2} & \dots & \frac{\partial f_m}{\partial x_n} \end{bmatrix} \begin{bmatrix} \Delta x_1 \\ \Delta x_2 \\ \vdots \\ \Delta x_n \end{bmatrix}$$

$$\underbrace{\left\{ \frac{\partial f_m}{\partial x_1}, \frac{\partial f_m}{\partial x_2}, \dots, \frac{\partial f_m}{\partial x_n} \right\}}_{\text{Jacobian matrix}} \begin{bmatrix} \Delta x_1 \\ \Delta x_2 \\ \vdots \\ \Delta x_n \end{bmatrix}$$

$$f(x+h) - f(x) \approx (\text{Derivative}) h$$

Case : $f: \mathbb{R} \rightarrow \mathbb{R}$

$$f(x+h) - f(x) \approx \underbrace{f'(x)} \underbrace{h}$$

$$f: \mathbb{R}^n \rightarrow \mathbb{R}$$

$$f(\underline{x} + \underline{h}) - f(\underline{x}) = \underline{\nabla f} \cdot \underline{h}$$

$$f: \mathbb{R}^n \rightarrow \mathbb{R}^m$$

$$\underline{f}(\underline{x} + \underline{h}) - \underline{f}(\underline{x}) = \underline{D}_f \underline{h}$$

Chain rule (general form).

$$f: \mathbb{R}^n \rightarrow \mathbb{R}^m$$

$$g: \mathbb{R}^k \rightarrow \mathbb{R}^n$$

$$(f \circ g): \mathbb{R}^k \rightarrow \mathbb{R}^m$$

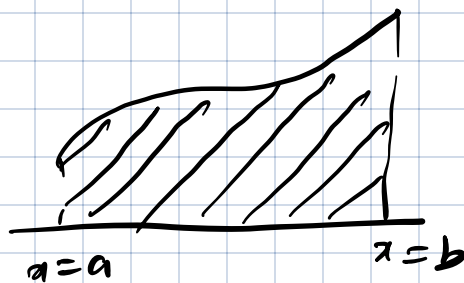
(let D_f be the Jacobian of f)

$$\boxed{f'(g(t)) g'(t)}$$

$$\underbrace{D_{f \circ g} = D_f(g) D_g}$$

Integrals

$f(x)$



$$\int_a^b f(x) dx$$

$$\boxed{F(y) = \int_a^y f(x) dx}$$

$$F'(y) = f(y)$$

Consider $f: \mathbb{R}^n \rightarrow \mathbb{R}$
(n variables)

$n=2$:

$$\iint_A f(x, y) dA$$

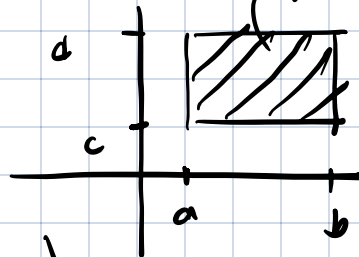
is referred to as the area
integral of f on A .

How to compute?

i) Rectangular regions:

$$\iint_A f(x, y) dA \quad \text{when}$$

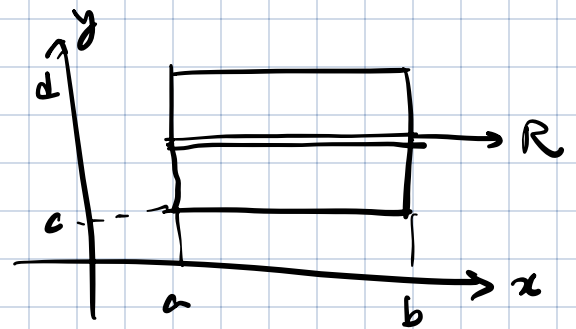
$$A = \left\{ \begin{array}{l} a \leq x \leq b \\ c \leq y \leq d \end{array} \right\}$$



(Fubini's theorem)

$$\iint_A f(x, y) dA = \int_{y=c}^{y=d} \left(\int_{x=a}^{x=b} f(x, y) dx \right) dy$$

Volume with
base R



$\iint_A (x + 2y) dA$ when A is rectangular region above.

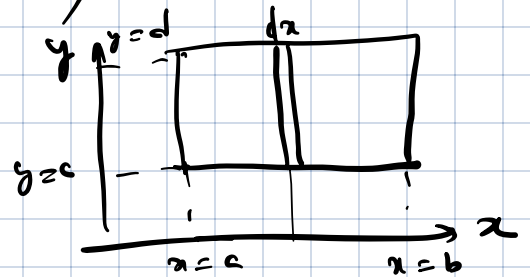
$$\left[\frac{x^2}{2} + 2xy \right]_{x=a}^{x=b}$$

$$\int_{y=c}^{y=d} \left(\frac{b^2 - a^2}{2} + 2y(b-a) \right) dy$$

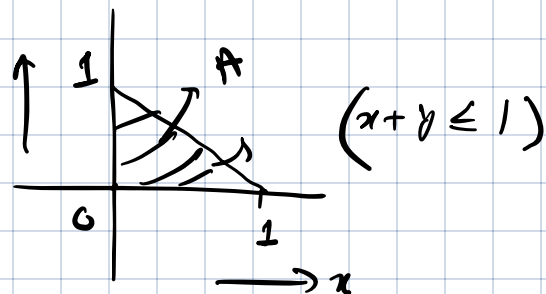
$$= \frac{b^2 - a^2}{2}(d-c) + \frac{2(b-a)(d^2 - c^2)}{2}$$

Likewise,

$$\iint f(x, y) dA = \int_{x=a}^{x=b} \left(\int_{y=c}^{y=d} f(x, y) dy \right) dx$$



2) More / general regions

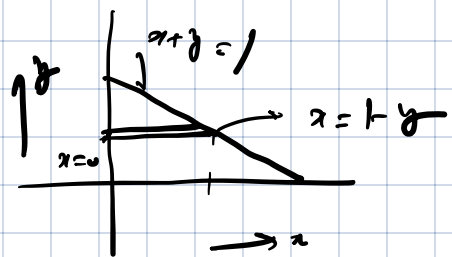


$$\iint_A f(x, y) dA$$

A: triangular region

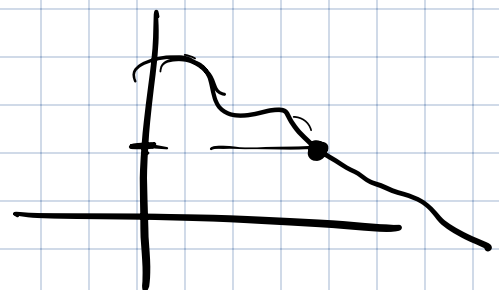
$$\int_{y=0}^1 \left(\int_0^{1-y} f(x, y) dx \right) dy$$

Type - II



Type - I

$$\int_{x=0}^1 \left(\int_0^{-x} f(x, y) dy \right) dx$$

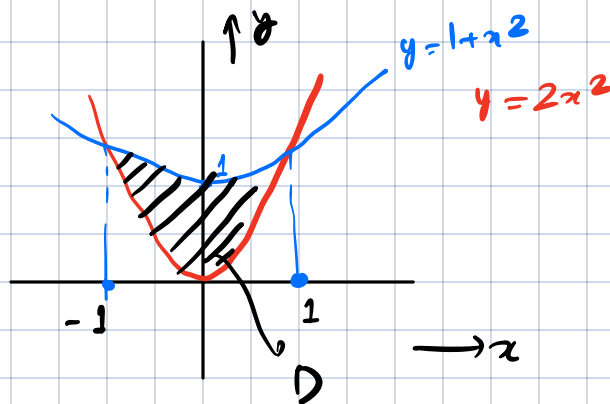


Exercise:

$$\iint_D (x+2y) dA$$

where D is the region enclosed

by the parabolas $y = 2x^2$, $y = 1 + x^2$.



$$2x^2 = 1 + x^2$$

$$x = \pm 1$$

for a particular x , y goes from $2x^2$ to $1 + x^2$

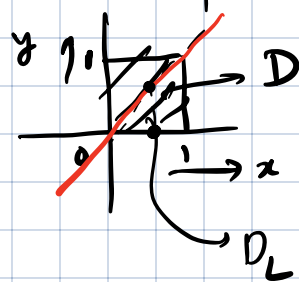
$$\int_{x=-1}^1 \left(\int_{2x^2}^{1+x^2} (x+2y) dy \right) dx$$

$$xy + y^2 \Big|_{y=2x^2}^{1+x^2}$$

Exercise :

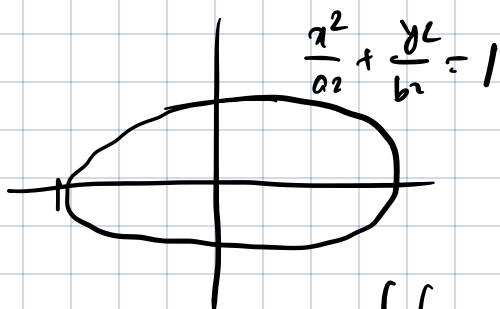
Find

$\iint_D \max\{x^2, y^2\} dA$ on the rectangular region



$$2 \iint_{D_L} x^2 dA$$

$$= 2 \int_{x=0}^1 \left(\int_{y=0}^x x^2 dy \right) dx$$

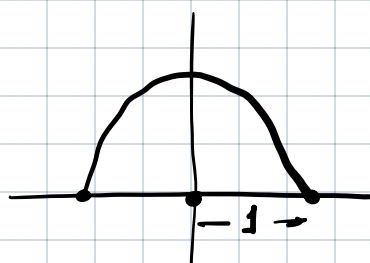


$$\iint_D dA = \text{Area of } D.$$

Center of mass:

$$\rho(x, y) = 1 + |xy|,$$

$$\iint \rho(x, y) dx dy = 1$$



simply connected regions

regions without "holes"

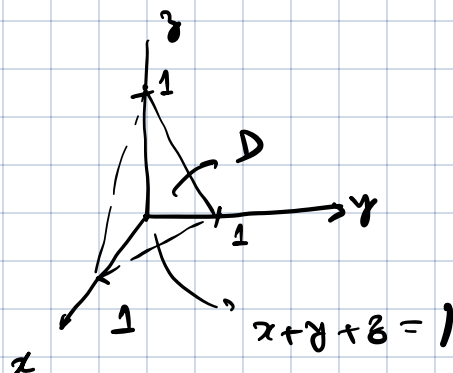
(x_c, y_c) be the center of mass

$$x_c = \iint_D x \rho(x, y) dA$$

$$y_c = \iint_D y \rho(x, y) dA$$



Beyond 2-variables, the principles of multiple integration are similar.



(Calculus: Early transcendentals)
J Stewart

$$\iiint_D dV = \int_0^1 \int_0^{1-x} \int_0^{1-x-y} f(x, y, z) dz dy dx$$

depend on x & y

depend on x

Exercise: Find the n -dimensional volume of the region

$$\{x_i \geq 0, \sum x_i \leq 1\} \subseteq \mathbb{R}^n.$$

Exercise: Find the volume of a 3-d sphere centered at origin & radius 1 by setting up the integral.

Change of variable in double integrals

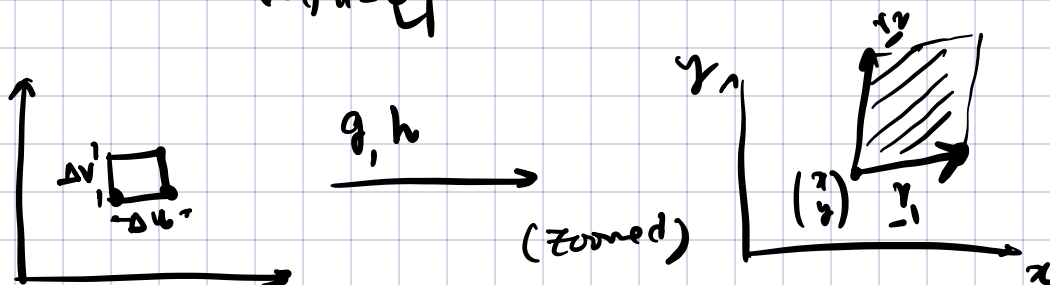
$$\iint_D f(x, y) dA \quad \begin{matrix} x = g(u, v) \\ y = h(u, v) \end{matrix}$$

$$= \int_{v=g}^{v=h} \int_{u=g}^{u=h} f(g(u, v), h(u, v)) du dv$$

$$\int f(x) dx$$

$$\int f(h(u)) h'(u) du$$

$$dx = h'(u) du$$



$$\begin{pmatrix} u \\ v \end{pmatrix} \rightarrow \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\begin{pmatrix} u + \Delta u \\ v \end{pmatrix} \rightarrow \begin{pmatrix} g(u + \Delta u, v) \\ h(u + \Delta u, v) \end{pmatrix}$$

Shaded

$$\text{area} = |\underline{r}_1 \times \underline{r}_2|$$

$$\underline{r}_1 = \begin{pmatrix} g(u + \Delta u, v) \\ h(u + \Delta u, v) \end{pmatrix} - \begin{pmatrix} g(u, v) \\ h(u, v) \end{pmatrix}$$

$$\begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ \frac{\partial g}{\partial u} & \frac{\partial h}{\partial u} & 0 \\ \frac{\partial g}{\partial v} & \frac{\partial h}{\partial v} & 0 \end{vmatrix} \Delta u \Delta v$$

$$= \begin{pmatrix} \frac{\partial g}{\partial u} \\ \frac{\partial h}{\partial u} \\ 0 \end{pmatrix} \Delta u$$

$$\underline{r}_2 = \begin{pmatrix} g(u, v + \Delta v) \\ h(u, v + \Delta v) \end{pmatrix} - \begin{pmatrix} g(u, v) \\ h(u, v) \end{pmatrix} = \begin{pmatrix} \frac{\partial g}{\partial v} \\ \frac{\partial h}{\partial v} \\ 0 \end{pmatrix} \Delta v$$

$$\begin{array}{c} dA \\ \text{area element} \\ \text{in } x, y \text{ plane} \end{array} = \left| \begin{array}{cc} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{array} \right| \underbrace{\Delta u \Delta v}_{\text{area element in } u, v \text{ plane.}}$$

$$dA = |\text{Jacobian}| \Delta u \Delta v$$

$$\left| \begin{array}{cc} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{array} \right|$$

Tool 1: $\iint_D f(x, y) dA = \iint_R f(x(u, v), y(u, v)) du dv |J|$

R : in u, v plane

D : in x, y plane

where $|J| =$ magnitude of determinant
of Jacobian

$$\left| \begin{array}{cc} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{array} \right|$$

under the following conditions

① $f, x(u, v), y(u, v)$ continuous, partial derivatives exist & continuous

② the map from $\begin{pmatrix} u \\ v \end{pmatrix} \rightarrow \begin{pmatrix} x \\ y \end{pmatrix}$ is one-one.

③ D is a simply connected region.

Example: Cartesian to polar.

$$\iint f(x, y) dx dy$$

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$D = \iint_R f(r \cos \theta, r \sin \theta) r dr d\theta \begin{vmatrix} \cos \theta & \sin \theta \\ -r \sin \theta & r \cos \theta \end{vmatrix} = r$$

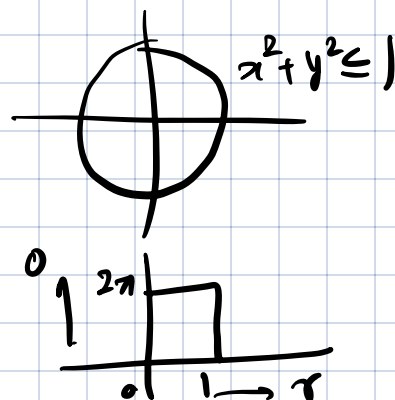
Exerc:

$$\iint_D \sqrt{x^2 + y^2} dx dy$$

D: unit circle
centred at origin.

$$= \int_{\theta=0}^{2\pi} \int_{r=0}^1 r^2 dr d\theta$$

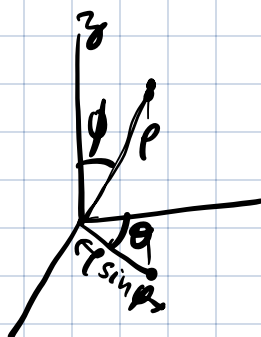
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Spherical co-ordinates

(x, y, z) cartesian

(ρ, ϕ, θ) spherical



$$z = \rho \cos \phi$$

$$x = \rho \sin \phi \cos \theta$$

$$y = \rho \sin \phi \sin \theta$$

Exercise Compute the det of Jacobian for this transformation.

Exercise Find $\iiint \sqrt{x^2 + y^2 + z^2} dV$

on the unit sphere centered on the origin.